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AMOROSILABS

TECHNICALBLUEPRINT · V1.0

Amorosi·HRM

Deterministic Cognitive Architecture

A neuro-symbolic system that treats hallucination as a translation error, not as a statistical artifact.

```
# amorosi.config
sycophancy: false
safe_narratives:
                    : false
deterministic: true
auditable:    true
```

AUTOR	FECHA	ESTADO	CLASIFICACIÓN
Juan P. Amorosi	February 2026	In Research	Blueprint

0 1 · THESIS

Hallucination is a translation error

LLMs are translators without a dictionary: they memorized statistical patterns between symbols without access to what those symbols mean.

Imagine a translator from Spanish to Japanese who never learned what the words mean. It only learned correlations: when "casa" appears, "家" tends to appear. With enough examples, it produces translations that seem correct. But ask it to translate "la casa del ser" in the Heideggerian sense, and it will produce something grammatically valid that speaks of real estate. It hallucinated — not because it is stupid, but because it never had access to meaning.

P R I N C I P I O F O U N D A T I O N A L

Hallucination is not a bug to minimize. It is a category error to eliminate. The solution is not more parameters — it is better grammar.

FORMALIZATION

Let f: Input → Output be a decision function. We require that there always exists a computable verification function:

$$\forall \text{ output} \in f(\text{input}): \exists\, g: \text{Output} \rightarrow \text{Verification where } g \text{ is computable} \wedge g(\text{output}) \text{ traces to evidence in Knowledge Graph } K$$

This is the Grounding Constraint: every element of the output must have a verifiable anchor. A hallucinating system produces syntactically valid outputs without there existing a g. A grounded system guarantees that g always exists and is traceable.

✗ Hallucinatory System

Handles form without access to content. Plausible output that may correspond to nothing real.

✓ Grounded System

Each symbol traceable to a finite vocabulary + evidence. Structural verification, not probabilistic.

The Operational Triad

Three capabilities that current systems dangerously confuse. They must never live in the same component: they have incompatible optimization objectives.

LAYER	RESPONSIBILITY	TECHNOLOGY	NATURE
Translation	Human ↔ machine interface	LLM	Fluid, adaptable
Execution	Logical reasoning	HRM	Rigid, deterministic
Memory	Knowledge grounding	Static graph	Immutable, auditable

P I P E L I N E E N R U N T I M E

DOORKEEPER

gateway

Schema validation (Pydantic), domain routing, rate limiting. No AI at the door — rejects invalid inputs instantly.

latency < 1ms · cost $0 · pure heuristics

↓

OPERATOR

encoder

Natural language → Grid 10×10. DeepSeek-8B with GRPO adapter. Constrained decoding: only tokens from the valid vocabulary.

latency ~200ms · cost ~$0 · deepseek-8b + lora

↓

BRAIN

hrm solver

Grid → Grid. Sapient HRM (27M params). Spatial pattern recognition — does not understand semantics, preserves structure.

latency ~100ms · cost $0 · sapient-hrm 27M

↓

SPEAKER

decoder

Grid → natural language. Synthesis with restricted grammar + injection of quotes from the Knowledge Graph. Only narrates what the grid sanctioned.

latency ~200ms · cost ~$0 · grammar-constrained

P R O P I E D A D C R Í T I C A

Semantic interpretation occurs only at the edges (encoder/decoder). Never at the core of reasoning. The HRM sees patterns, not meanings.

0 3 · PRINCIPIO CENTRAL

Semantic-Numeric Isomorphism

Every decision problem can be mapped to a spatial topology with finite vocabulary. Verification occurs by topological constraints, not by probabilistic confidence.

POSITIONAL GRAMMAR · GRID 10×10

The position encodes the semantic role. The value encodes the primitive. Semantics emerge from the combination.

cols 0–2 · current state
cols 3–5 · transformation
cols 6–9 · target state

rows 0–2 · primary entities
rows 3–5 · relationships
rows 6–9 · conditions / context

EXAMPLE · RETAIL PRICING

1	0	0	0	0
9	0	0	0	0
5	0	0	0	0
·	·	·	·	·
?	0	0	0	0

- row 0 · INVENTORY = 1 (crit-low)
- row 1 · DEMAND = 9 (crit-high)
- row 2 · COMPETITION = 5 (normal)
- row 5 · DECISION = ? (to be resolved)

WHAT THE HRM REALLY SEES

The HRM does not "understand" that row 0 is inventory. It sees a spatial pattern: obstacle(1) + pressure(9) + neutral(5), and produces a learned output pattern.

```
# output
row 5 · DECISION = 12       / INC_HARD → +15% price increase
```

Why it eliminates hallucination: when the output space is closed and enumerated, hallucinating a non-existent primitive is structurally impossible. There is no room to invent meaning.

04 · EXTENSIBILITY

The Cartridge Model

Domain knowledge is encapsulated in hot-swappable cartridges—like console cartridges. One kernel, many domains.

ANATOMY OF A CARTRIDGE

```
retail_pricing/
├── vocabulary.json # primitives
├── rules.yaml # constraints
└── weights.safetensors # LoRA
```

PLANNED CARTRIDGES

retail_pricing · logistics_routing · credit_scoring · medical_triage · financial_compliance

CARTRIDGE LIFECYCLE

PHASE	PROCESS	OUTPUT
DEFINITION	Vocabulary + rules + business constraints	schema
COMPILATION	GRPO training on synthetic + real data	LoRA adapter
SIGNING	Cryptographic hash of all components	signature
RUNTIME	Hot-swap mounted in the kernel	live

ARCHITECTURAL ADVANTAGE

A new domain does not require re-training the reasoning engine. Just compile a new cartridge. The kernel is immutable; knowledge is modular.

Contrastive Predictive Cartridge Idea2Paper evolution

The academic iteration extends the model: cartridges not only contain static knowledge, but learn to predict their own evolution through contrastive modeling of future states, with cryptographic integrity in signed predictive logs. Adaptation shifts from reactive correction to anticipatory coherence.

GRPO & Deterministic Reward

Group Relative Policy Optimization: use the group mean as the baseline. No separate Critic, real gradients, efficient for deterministic reward functions.

WHY GRPO VS ALTERNATIVES

METHOD	LIMITATION
PPO	Critic = 2× VRAM
DPO	Only win/lose, without gradients
GRPO	No Critic, real gradients

TRAINING LOOP

1. Generate G=16 grids per case
2. Evaluate reward (Rust)
3. baseline = mean(rewards)
4. advantage = R − baseline
5. Reward if adv > 0

REWARD FUNCTION · THE HEART OF THE SYSTEM

```
def amorosi_reward(grid, vocabulary, grammar, context):
    reward = 0.0

    # 1 · GRAMMAR CHECK — terminal penalty
    for rule in grammar.rules:
        if not rule.check(grid): return -5.0
    reward += 1.0

    # 2 · VOCABULARY CHECK — hallucinated primitive = punishment
    for cell in grid.flatten():
        if cell not in vocabulary.valid: return -3.0
    reward += 1.0

    # 3 · BUSINESS LOGIC — simulation of outcome [0.0–2.0]
    reward += simulate_outcome(grid, context)

    # 4 · PARSIMONY — prefer simple solutions
    reward -= np.count_nonzero(grid) * 0.01

    return reward
```

The reward function is the system's definition of "truth." Written in Rust, it runs millions of times during training: it must be fast and unambiguous. The model explores multiple solutions and converges to the optimum.

0 6 · INFERENCE & ECONOMY

Cascade & the Data Flywheel

All that can be computed offline should be computed offline. The factory is capital investment; the store has near-zero marginal cost.

LEVEL 0 · Rust / HRM — \$0

Direct lookup in pre-computed decision tables. Queries pattern-matchable.

↓ if not in table

LEVEL 1 · 8B + GRPO (local) — ~\$0.001

Complete pipeline encode → HRM → decode. Covers ~90% of the queries.

↓ if confidence drops

LEVEL 2 · V3 / R1 (cloud) — ~\$0.05

Full cloud inference + capture for retraining. Edge cases (~10%).

DATA FLY WHEEL

Each Level 2 solution becomes training data for Level 1. The system improves continuously — today's expensive queries are free tomorrow.

COMPARATIVE ECONOMICS

METRIC	GPT-4 WRAPPER	AMOROSI-HRM
Cost / query	\$0.03–0.10	~\$0.001
Latency	5–30s	0.5–1s
Hallucination	~15%	<0.1% *
Auditability	None	Complete

* design goal — hypotheses to validate empirically, not measured result

07 · INTERFAZ

Six Cognitive Endpoints

The API exposes mental operations, not business domains. A cognitive kernel, analogous to the kernel of an operating system.

POST	/decide	Optimal selection under constraints — pricing, credit approval
POST	/plan	Temporal sequencing of states — routing, scheduling
POST	/diagnose	Inverse causal inference — medical triage, debugging
POST	/verify	Constraint satisfaction auditing — compliance, fact-check
POST	/classify	Ontological mapping — ticket classification, NER
POST	/simulate	Projection of future states — risk, what-if

PROTOCOLSANTI-ALUCINACIÓN

PROTOCOL	MECHANISM	LOCATION
Constrained Decoding	Logit bias enforce valid tokens	Encoder
Grammar Loss	Penalty for topology violation	GRPO
Mirror Check	Deterministic Parser checks Grid↔Text	Post-Decoder
Citation Audit	References validated against the graph	Output

TWO-LAYER RESPONSE

proof_layer — verifiable and persistent: grid_hash, activated path, citations, confidence. view_layer — cosmetic and disposable: legible decision, explanation. The evidence never depends on narration.

0 8 · L I N A J E & E S T A D O

From the Hittites to the tensors

The Hittites adapted Sumerian cuneiform preserving meaning between cultures. They translated form preserving function. Juan's father, a language historian, studied how symbols migrate and how grammar evolves. This project is the same in another substrate.

INTELLECTUAL HERITAGE

Hans Uszkoreit (linguist)
→ Jakob Uszkoreit (Transformer)
→ attention = relationship between symbols

Father Amorosi (historian)
→ Juan Amorosi (grounding)
→ anchoring to verifiable meaning

THE DANCER

Gears are only numbers. The dancer is only interpretation. The magic is that both things are true simultaneously.

STATE ACTUAL & ETHIC OF PISTEMICA

✓ Completed

- Foundational document v0.1
- Blueprint v1.0
- GRPO mechanism specified
- Reward function defined
- API spec · 6 endpoints
- Whitepaper Idea2Paper

○Pending

- Executable MVP
- 10k synthetic cases
- Train GRPO adapter
- Integrate Sapient HRM
- Measure real GTCR
- Empirical validation

P O S I T I O N H O N E S T A

Amorosi-HRM is an ambitious architectural hypothesis in active development — not a proven system. Metrics are design objectives, not measured results. That honesty is the strength, not the weakness.

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AMOROSILABS

"Anchor decisions to verifiable structure."

DETERMINISTIC · AUDITABLE · SOVEREIGN